Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
13	*(a)*			Elfed's reaction produces Cr^{2+} (1) dichromate(VI) can be reduced to Cr^{3+} and further to Cr^{2+} as E^{θ} values for these two processes are more positive than that for Zn^{2+}/Zn / EMF values for reactions are positive (2.09 V and 0.34 V) (1) Fatima's reaction produces Cr^{3+} (1) oxygen oxidises Cr^{2+} back to Cr^{3+} as E^{θ} value is more positive than that for Cr^{3+}/Cr^{2+} / EMF value for reaction is positive (0.82 V) (1)			1 1 1 1	4		

Question				Marking details	Marks available AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		$K_c = \frac{[CrO_4^{2-}]^2[H^+]^2}{[Cr_2O_7^{2-}]\ \ [H_2O]}$		1		1		
		(ii)		1 dm^3 contains 1000g of water (1) $\frac{1000}{18.02}$ = 55.5 in 1 dm^3 (1)	1	1		2		
		(iii)		$[H^+]^2 = \frac{Kc \times [Cr_2O_7^{2-}] \times [H_2O]}{[CrO_4^{2-}]^2}$ (1) $[H^+]^2 = 2.356 \times 10^{-12}$ or $[H^+] = 1.535 \times 10^{-6}$ mol dm^{-3} (1) pH = 5.81 (1) student is incorrect as this solution is acidic (1)		1	1 1 1	4	3	
		(iv)		if K_c increases the reaction is shifting to the right (1) if a reaction shifts to right when heated it must be endothermic (1)		1	1	2		
				Question 13 total	**1**	**4**	**8**	**13**	**3**	**0**